RATING METHODOLOGY

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TABLE OF CONTENTS

Scope	1
Sector overview	2
Rating approach	3
Scorecard	4
Discussion of the scorecard factors	6
Other considerations	10
Scorecard-indicated outcome	12
Assigning issuer-level and instrument-level ratings	13
Key rating assumptions	14
Limitations	14
Moody's related publications	15

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Rating Methodology

Communications Infrastructure

This rating methodology replaces the *Communications Infrastructure* methodology published on 25 February 2022. We have made changes to the scorecard to provide greater consistency of approach across our corporate sector methodologies. In the Business Profile factor, we have renamed the Competitive Environment and Business Conditions subfactor to Market Position and Market Characteristics. In the Profitability and Efficiency factor, we have replaced the FFO Margin subfactor with EBITDA Margin. In Leverage and Coverage, we have replaced the FCF/Debt subfactor with RCF/Net Debt. In addition, we have changed some scoring thresholds. We have also made editorial changes throughout the methodology to enhance readability, and we have changed the presentation of the scorecard.

Scope

This rating methodology applies to companies globally that are primarily engaged in providing communications infrastructure. Communications infrastructure companies provide specific physical or network infrastructure, including data transport services or processing capacity, to telecommunications carriers, pay television companies, corporations and government entities, as well as other types of companies and institutions.

The determination of a company's primary business is generally based on the preponderance of the company's business risks, which are usually proportionate to the company's revenue, earnings and cash flows.

Although communications infrastructure companies provide telecommunications services, they are generally more specialized than the companies that we rate using our methodology for telecommunications service providers, and they provide communications infrastructure on a wholesale basis rather than to retail consumers directly.[1]

We rate tower companies and data centers that are or convert to real estate investment trusts (REITs) using our methodology for REITs and other commercial real estate firms. We also use our methodology for REITs and other commercial real estate firms to rate other companies involved in the communications infrastructure business that are not REITs. Such companies encompass those whose primary business includes significant ownership, development or management of real estate properties, projects or companies, and those whose income stream is principally derived from leases of real estate assets or other real estate-based activities, rather than from activities tied to technological equipment and associated risks.

This rating methodology was republished on 7 November 2025. We corrected an error in the Ca threshold for the (EBITDA - CAPEX) / Interest Expense subfactor in the scorecard.

Sector overview

Communications infrastructure companies typically offer their customers a lease-like, off-balance-sheet alternative to owning and financing the assets themselves. The physical assets that communications infrastructure companies own play an essential role in delivering mobile phone service, internet capacity and broadcast television to consumers. The industry is capital intensive, and companies require financial resources in order to support the large upfront investments needed for growth.

For example, a fiber network company might receive a significant upfront cash payment for the sale of dark fiber (unlit fiber), and then amortize the revenue over the 10 years the customer will use that dark fiber. The company will continue to record revenue and EBITDA, typically at a very high margin, over the life of the contract, but it will not receive additional cash payments. This non-cash EBITDA strengthens the debt-to-EBITDA ratio and interest-coverage metrics but does not fund debt repayment or interest expense. The free-cash-flow metric, on the other hand, reflects when the company received the cash.

Another potentially important mismatch arises when data center operators have contracts that contain rent escalators. A data center company often pays lower rent on a cash basis in the early years of a lease, which helps it achieve profitability on a cash basis as it secures more clients to fill its space.

Nearly all companies in the sector benefit from some degree of recurring revenue and long-term contracts, as well as location exclusivity that creates barriers to entry.

In terms of business models, we categorize the companies that we rate using this methodology as satellite operators; independent tower operators that are independent of wireless telecommunications service providers, which sometimes operate their own towers; or fiber networks and data centers.

» Satellite operators provide their services using geostationary earth orbit (GEO) satellites, medium Earth orbit (MEO) satellites, low Earth orbit (LEO) satellites or a combination of these satellite types. The choice of satellite type depends on the specific requirements and limitations of the application. GEO satellites are preferred for applications that require continuous coverage over a specific region, while LEO satellites are preferable for low latency communications. MEO satellites fall in-between. All three satellite types are vital for communication, based on their orbital characteristics.

» Fixed satellite service operators, which constitute the majority of the rated operators, transmit communications between stations on the ground and GEO satellites, which move with the earth and appear to maintain a fixed point in the sky. These satellites can offer a cost-effective way of broadcasting the same information to multiple recipients across a wide geography. For example, the broadcast of a linear television channel (whereby a broadcaster sets the programming schedule, as opposed to on-demand television) usually relies on satellite service. Although satellites are generally not as cost effective as terrestrial networks for two-way communication, they remain indispensable for sparsely populated, remote or maritime markets.

» Wireless tower operators provide infrastructure and services that enable wireless carriers to provide last-mile communications to end users. Last-mile communications refers to technology that is used to connect physical and network infrastructure to a home or business. These operators generally have very long-term contracts with telecomunications companies, which allows for a high degree of revenue visibility and predictability. Therefore, wireless tower operators can operate with more leverage compared with other companies in the communications infrastructure industry.

Fiber networks provide the backbone for internet connectivity. Data centers provide customers with dedicated sites that house data processing and network equipment and maintain redundant network access in a controlled environment with high security at a lower cost than customers could get investing in the infrastructure themselves. Data center products may also include managed services, bandwidth services and security services.

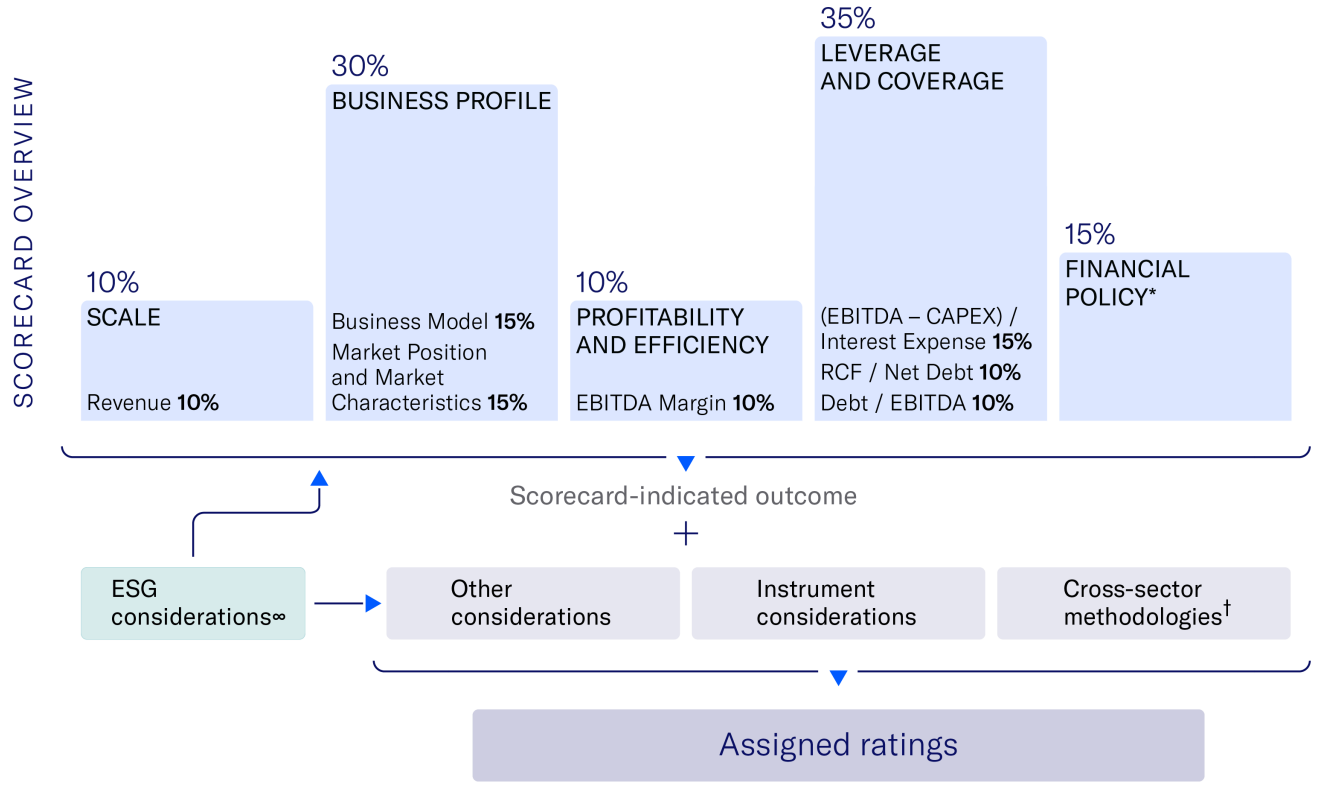
Rating approach

In this methodology, we explain our general approach to assessing credit risks of companies in this sector, including the quantitative and qualitative factors likely to affect rating outcomes.

The following exhibit illustrates our overall rating approach in the analysis of companies in this sector, which includes the use of a scorecard. The scorecard-indicated outcome may not match a company's assigned rating, which is determined by a rating committee. For more information, see the "Other considerations" and "Limitations" sections.

Exhibit 1

Illustration of the communications infrastructure rating approach



* This factor has no subfactors.

∞ Environmental, social and governance (ESG) considerations, including, where available, our opinions of exposure to them as expressed in Issuer Profile Scores (IPSs), may affect scorecard factors and other considerations outside of the scorecard. For more information, see the "Other considerations" section.

† Some of the methodological considerations described in one or more cross-sector methodologies may be relevant to ratings in this sector. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

Source: Moody's Ratings

Scorecard

The scorecard in this methodology is a relatively simple reference tool that can be used in most cases to approximate credit profiles of companies in this sector and to explain, in summary form, many of the factors that are generally most important in assigning ratings. The scorecard may not include or address every consideration a rating committee may assess in assigning ratings to companies in this sector.

Exhibit 2

Communications infrastructure scorecard

	Weight	Aaa	Aa	A	Baa	Ba	B	Caa	Ca
Factor: Scale (10%)									
Revenue (USD Billion)[1]	10%	≥ $40	$20 - $40	$10 - $20	$4 - $10	$1.5 - $4	$0.5 - $1.5	$0.25 - $0.5	<$0.25
Factor: Business Profile (30%)									
Business Model	15%	**Satellite**: Global FSS provider that owns a very large constellation of satellites, in-orbit redundancy, exceptional customer diversity, and extremely broad industry exposure; exclusive orbital slots to provide coverage around the world, including significant spectrum capacity to handle high-bandwidth data transmissions; other license awards are prohibited by regulators; and in-market fiber capacity is non-existent. **Towers:** Dominant global provider of independent tower sites to wireless carriers in regions where the operator controls substantially all tower sites, and government regulations prohibit new tower construction; operator owns the towers and land; and customers are spread across the wireless carriers' markets. **Fiber Networks and Data Centers**: Dominant global provider of data center services, fiber connectivity (with broad product diversity) or network interconnection with exclusive internet access points in its service area; exceptional client diversity; company owns all of the data center properties or fiber routes it serves; has primary leasehold rights or easements in buildings and floors/risers it serves; and regulations prevent entry of new competitors.	**Satellite**: Global or regional FSS provider that owns a large constellation of satellites, in-orbit redundancy, excellent client diversity, and very broad industry exposure; exclusive orbital slots to provide coverage around the world, including significant spectrum capacity to handle high-bandwidth data transmissions; and in-market fiber capacity is non-existent. **Towers**: Multi-national provider of independent tower sites to wireless carriers in regions with developed independent tower sector in which independent owners control greater than a majority of industry cell sites; location exclusivity and zoning regulations prohibit new tower construction; and customers are spread across the wireless carriers' markets. **Fiber Networks and Data Centers**: Dominant global or national provider of data center services, fiber connectivity (with broad product diversity) or network interconnection with exclusive internet access points in its service area; excellent customer diversity; company owns all of the data center properties or fiber routes it serves; has primary leasehold rights or easements in buildings and floors/risers it serves; and regulations limit entry of new competitors.	**Satellite**: Global or regional FSS provider, with a large constellation of satellites, in-orbit redundancy, very good client diversity, and broad industry exposure; desirable orbital slots to provide coverage around the world, including significant spectrum capacity to handle high-bandwidth data transmissions. **Towers**: Multi-national provider of independent tower sites to wireless carriers in regions with developed independent tower sector in which independent owners control at least a majority of the cell sites; location exclusivity and zoning regulations significantly constrain new tower construction; customers are spread across the wireless carriers' market. **Fiber Networks and Data Centers**: Global or national provider of data center services, fiber connectivity (with broad product diversity) or network interconnection at critical internet access points, resulting in location exclusivity; very good customer diversity; company owns the vast majority of the data center properties or fiber routes it serves; has primary leasehold or easements in buildings and floors/risers it serves.	**Satellite**: Regional FSS provider with a medium-size constellation of satellites, good client diversity and broad industry exposure; desirable orbital slots to provide coverage in the regions it serves. **Towers**: National or regional provider of independent tower sites to wireless carriers in regions with developed independent tower sector in which independent owners control a meaningful amount of the cell sites; location exclusivity and zoning regulations constrain new tower construction; some customer concentration exists, albeit with a dominant carrier. **Fiber Networks and Data Centers**: National or regional provider of data center services, fiber connectivity (with broad product diversity) or network interconnection at critical internet access points; good customer diversity; company owns a majority of the fiber routes it serves; has primary leasehold or easements in buildings and floors/risers it serves.	**Satellite**: Single-country FSS provider in area with increasing fiber capacity, or global VSAT[6] provider with good client diversity but lacks broad product diversity (primarily data services to maritime and remote regions); limited owned transponder capacity, but has relationships with major FSS providers to provide coverage around the world. **Towers**: Regional provider of independent tower sites to wireless carriers in regions with less-developed independent tower sector in which wireless carriers control more than a majority of the cell sites; limited zoning restrictions on new tower construction. Some customer concentration exists, albeit with a dominant carrier. **Fiber Networks and Data Centers**: Regional provider of data center services, fiber connectivity or network interconnection; good customer diversity, but lacks broad product diversity; company leases most of the fiber routes and properties it serves; relies on secondary arrangements to gain access into buildings and floors/risers it serves.	**Satellite**: Single-country FSS provider in area with significant, built-out fiber capacity, or regional VSAT provider with significant customer concentration, limited client diversity and lack of broad product diversity (primarily data services to maritime and remote regions); limited owned transponder capacity. **Towers**: Regional provider of independent tower sites to wireless carriers in regions with less-developed independent tower sector in which wireless carriers control more than the majority of cell sites; few zoning restrictions govern new tower construction; customer concentration exists. **Fiber Networks and Data Centers**: Limited-reach regional provider of data center services, fiber connectivity or network interconnection; limited customer diversity and lacks broad product diversity; company leases most of the fiber routes and properties it serves; lacks significant access to buildings and floors/risers in markets it serves.	**Satellite**: Primarily a reseller of spare satellite capacity; aging fleet of satellites requiring significant replacements; or limited bandwidth capacity. **Towers**: Local provider of independent tower sites to wireless carriers in regions where the independent tower sector is in the very early stages of development; few zoning restrictions govern new tower construction; or significant customer concentration exists. **Fiber Networks and Data Centers**: Local provider of data center services, fiber connectivity or network interconnection; lacks both broad product and customer diversity; company leases substantial portions of its fiber routes and the properties it serves; or lacks significant access to buildings and floors/risers in markets it serves.	**Satellite**: Primarily a reseller of spare satellite capacity; dependent on a single satellite needing replacement in the near term, for which no committed funding source exists; or limited bandwidth capacity. **Towers**: Start-up provider of independent tower sites to wireless carriers in regions where the independent tower sector is not developed; no zoning restrictions to constrain new tower construction; or significant customer concentration with weak wireless services providers exists. **Fiber Networks and Data Centers**: Very small provider of data center services, fiber connectivity or network interconnection; lacks broad product diversity and has high customer concentration; company leases substantially all of the fiber routes and the properties it serves; or lacks significant access to buildings and floors/risers.

	Weight	Aaa	Aa	A	Baa	Ba	B	Caa	Ca
Market Position and Market Characteristics	15%	Exclusive provider of services to its clients within the coverage area; very long-term contracts; unique service offering unmatched by peers; virtually no customer churn; and government regulations restrict virtually any entry of new competitors.	Dominant provider of services to its clients within the coverage area; very long-term contracts; highly specialized service offering; very low customer churn; and government regulations severely limit entry of new competitors.	Strongly competitive provider of services within the coverage area; long-term contracts; specialized service offering; low customer churn; high barriers to entry for competitors and high barriers to exit for the customer.	Competitive provider of services within the coverage area against a modest number of viable alternative providers; long-term contracts; some service differentiation; modest customer churn; high barriers to exit for the customer.	Reasonably competitive provider within the coverage area against some viable alternative providers; primarily medium-term contracts; limited differentiation of service offering; moderate customer churn; few barriers to entry other than availability of capital; high barriers to exit for the customer.	Modestly competitive provider, or industry has many viable alternatives to the service the company provides; primarily short-term contracts; minimal differentiation of service offering; meaningful customer churn; barriers to entry limited to availability of capital; high barriers to exit for the customer; faces competition from IT integrators and telcos.	Weakly positioned provider or industry has a large number viable, competitive alternatives to the service the company provides; short-term contracts; commodity-like service offering; high customer churn; barriers to entry limited to availability of capital; low barriers to exit for the customer; or faces competition from IT integrators and telcos.	Very weakly positioned provider, or industry has a great many viable, highly competitive alternatives to the service the company provides; or short-term contracts with weak customers; commodity-like service offering; very high customer churn; minimal entry barriers for competitors and low barriers to exit for customers; or faces high competition from IT integrators and telcos.
Factor: Profitablity and Efficiency (10%)									
EBITDA Margin [2]	10%	≥ 100%	80% - 100%	70% - 80%	60% - 70%	45% - 60%	30% - 45%	20% - 30%	< 20%
Factor: Leverage and Coverage (35%)									
(EBITDA - CAPEX) / Interest Expense [3]	15%	≥ 12x	7x - 12x	4x - 7x	3x - 4x	2x - 3x	1x - 2x	0.5x - 1x	< 0.5x
RCF / Net Debt [4]	10%	≥ 80%	60% - 80%	35% - 60%	25% - 35%	15% - 25%	10% - 15%	5% - 10%	< 5%
Debt / EBITDA [5]	10%	≤ 0.5x	0.5x - 1.0x	1.0x - 2.0x	2.0x - 3.0x	3.0x - 4.5x	4.5x - 6.5x	6.5x - 8.5x	>8.5x
Factor: Financial Policy (15%)									
Financial Policy	15%	Expected to have extremely conservative financial policies (including risk and liquidity management); very stable metrics; essentially no event risk that would cause a rating transition; and public commitment to a very strong credit profile over the long term.	Expected to have very conservative financial policies (including risk and liquidity management); stable metrics; minimal event risk that would cause a rating transition; and public commitment to a strong credit profile over the long term.	Expected to have predictable financial policies (including risk and liquidity management) that preserve creditor interests; although modest event risk exists, the effect on leverage is likely to be small and temporary; strong commitment to a solid credit profile.	Expected to have financial policies (including risk and liquidity management) that balance the interests of creditors and shareholders; some risk that debt-funded acquisitions or shareholder distributions could lead to a weaker credit profile.	Expected to have financial policies (including risk and liquidity management) that tend to favor shareholders over creditors; above-average financial risk resulting from shareholder distributions, acquisitions or other significant capital structure changes.	Expected to have financial policies (including risk and liquidity management) that favor shareholders over creditors; high financial risk resulting from shareholder distributions, acquisitions or other significant capital structure changes.	Expected to have financial policies (including risk and liquidity management) that create elevated risk of debt restructuring in varied economic environments.	Expected to have financial policies (including risk and liquidity management) that create elevated risk of debt restructuring even in healthy economic environments.

[1] For the linear scoring scale, the Aaa endpoint value is $60 billion. A value of $60 billion or better equates to a numeric score of 0.5. The Ca endpoint value is zero. A value of zero equates to a numeric score of 20.5.
[2] For the linear scoring scale, the Aaa endpoint value is 120%. A value of 120% or better equates to a numeric score of 0.5. The Ca endpoint value is 10%. A value of 10% or worse equates to a score of 20.5.
[3] For the linear scoring scale, the Aaa endpoint value is 18x. A value of 18x or better equates to a numeric score of 0.5. The Ca endpoint value is zero. A value of zero or worse equates to a numeric score of 20.5.
[4] For the linear scoring scale, the Aaa endpoint value is 100%. A value of 100% or better equates to a numeric score of 0.5. The Ca endpoint value is zero. A value of zero or worse equates to a numeric score of 20.5. When net debt is negative and RCF is positive, the numeric score is 0.5. When net debt is negative and RCF is negative or zero, the numeric score is 20.5.
[5] For the linear scoring scale, the Aaa endpoint value is zero. A value of zero equates to a numeric score of 0.5. The Ca endpoint value is 12x. A value of 12x or worse equates to a numeric score of 20.5, as does a negative Debt/EBITDA value.
[6] Very Small Aperture Terminals (VSAT) are small earth stations that share satellite resources among a large number of similar terminals. Individual VSAT terminals typically transmit at relatively low power levels, and use relatively small equipment that allows flexible installation of a satellite network earth station directly at a wide variety of user locations and platforms.
Source: Moody's Ratings

Discussion of the scorecard factors

In this section, we explain why each scorecard factor or subfactor is a meaningful credit indicator, and we discuss our analytical approach to scoring each factor or subfactor. When a factor comprises subfactors, we score at the subfactor level. Some factors do not have subfactors, in which case we score at the factor level.

In assessing quantitative subfactors, we use information that we calculate or estimate from a company's financial statements or regulatory filings, or that we derive from other observations. We may incorporate nonpublic information qualitatively in our assessment.

Our ratings are forward-looking and reflect our expectations for future financial and operating performance. However, historical results are helpful in understanding patterns and trends in a company's performance, as well as for peer comparisons. We typically calculate or estimate financial metrics, unless otherwise indicated, based on an annual or other 12-month period. However, we may assess the scorecard factors using various periods. For example, we may find it analytically useful to examine both historical and expected future performance for periods of several years or more.

All of the quantitative credit metrics in the scorecard incorporate our standard adjustments to income statement, cash flow statement and balance sheet amounts for items described in our cross-sector methodology on financial statement adjustments for nonfinancial corporations.[2] We may also make nonstandard adjustments that are specific to a particular company.

For a discussion of scorecard mechanics, see the "Scorecard-indicated outcome" section.

Factor: Scale (10% weight)

Why it matters

Scale is an important indicator of the overall strength of a company's business, the size of its market and the company's relevance within an industry and the economy. Scale is also an indicator of a company's importance to its customers and its bargaining strength with suppliers, as well as its ability to invest resources to maintain its competitive position and expand its marketing and geographic reach.

How we assess it for the scorecard

We score this factor based on one quantitative subfactor: Revenue.

Revenue

We use total reported revenue in billions of US dollars to measure scale, or to estimate it in the case of forward-looking expectations.

Factor: Business Profile (30% weight)

Why it matters

The business profile of a company is important because it greatly influences its ability to generate sustainable earnings and operating cash flow.

In this sector, business profile provides critical indications of the strength or weakness of a company's business model based on whether the company owns or leases assets, its product and service offerings, and the exclusivity of the location of its infrastructure.

A communications infrastructure company's competitive environment and business conditions, including the stability and tenor of its contracts, also affects its ability to generate cash flow and raise external capital. All else equal, barriers to entry and long-term contracts with strong counterparties can foster stable cash flows that may support higher leverage than companies that face more robust competition.

Business Model

Most communications infrastructure providers are asset heavy. Asset ownership is a credit strength because owning the assets gives an operator more control over its costs than reselling the spare capacity of other providers. Also, companies need some excess asset capacity to ensure as close to zero downtime as possible for customers, given the critical nature of services provided. For example, satellite companies need redundancy to manage possible in-orbit malfunctions.

A communications infrastructure provider with expansive geographic reach and a broad product line is able to serve a diverse customer base, which helps insulate it from region-, industry- or customer-specific weaknesses. Tower companies, which focus on serving wireless operators, tend to have a less diversified customer base than other types of communications infrastructure providers; however, high barriers to entry, the essential nature of the service they provide and the typically strong credit profiles of their customers help offset this risk.

In addition to ownership and breadth of assets, location exclusivity can create barriers to entry. For example, the high cost of building out a fiber network makes overbuilding less likely where there are already alternative providers, and a company with an expansive network of fiber routes can protect its position. Similarly, data center operators with facilities in key network locations have a greater opportunity to interconnect with other carriers or customers, a market advantage that may be difficult for other operators to replicate.

Market Position and Market Characteristics

The market position and characteristics of a company are important since they directly affect its pricing power and marketing expenses. Communications infrastructure companies often offer services that their key customers could self-provide (if they were willing to make the required investments). Therefore, reliability, product differentiation, execution and competitive cost structures are critical. A product set that includes exclusive, specialized offerings leads to pricing power, whereas more commodity-like products and services could make a company susceptible to supply-demand imbalances and volatile service pricing. A differentiated offering also usually leads to higher switching costs for customers, which can lend stability to the customer base. The ability to innovate and keep pace with technological change is necessary for a company to maintain market share and achieve growth.

A stable customer base, long-term contracts and low customer churn are often essential to ensuring an adequate return on investment given the high capital intensity in the sector, especially since companies typically invest capital before they start to generate cash flow from their contracts.

Customer churn can have varying degrees of negative impact for the different subsectors. In some cases, such as satellite or tower operators, some churn is positive in that it allows for price escalation in markets where demand is strong. However, fiber carriers that focus on local connectivity are unlikely to be able to reuse equipment, and high churn can have disproportionately negative effects on cash flow.

How we assess it for the scorecard

We score this factor based on two qualitative subfactors: Business Model, and Market Position and Market Characteristics.

Business Model

For all subsectors, we compare the proportion of assets that the company owns to the proportion it leases. For example, a satellite operator with a large constellation of owned satellites will typically score better for this subfactor than one that resells the spare capacity of other operators.

For a data center or fiber network operator, we typically consider the ownership status of the network it uses to serve multiple customers and how it accesses the buildings of specific customers. A data center or fiber network operator that owns significant interconnection facilities in locations critical to customers will generally receive a higher score for this subfactor than one that leases more of its assets and relies on secondary arrangements for access to the buildings of the customers it serves.

We assess diversification across customers and geographic reach. We typically consider the number of total customers and revenue concentration in customers or industries, based on available data. For tower companies, which focus on serving wireless operators and tend to have a less diversified customer base, we may assess whether the revenue is spread across the major wireless operators as well as the financial profile of those customers, because strong wireless operators themselves typically have highly diversified and more stable customer bases.

For satellite operators and fiber networks and data centers, we consider both the breadth of the industries they serve and the number of customers. We view international diversification positively if it affords the provider with key strategic landing points or internet gateways in regions with high traffic. We view it negatively if the diversification is in emerging markets and likely to add risk. The

location of a company's assets generally provides some indication of its customer base, although we may consider any segment disclosure in financial reporting.

Our assessment of location exclusivity can vary by subsector, but we typically consider the size and location of the asset base. Satellite operators can achieve an exclusive market presence with orbital slots and spectrum capacity, and we typically look at the location, number and redundancy of satellites.

For tower operators, we typically consider the number and location of its towers. Barriers to entry can result from zoning regulations, and tower operators that own assets in markets with heavy zoning restrictions generally score higher for this subfactor.

For fiber networks, we typically consider the length and location of a company's network; for example, we may assess the route miles or kilometers of a fiber network. For a data center operator, we consider the number and locations of data centers and the total floor space or power capacity of its facilities.

Market Position and Market Characteristics

We typically assess a company's market position and market characteristics based on the number and strength of competitors that provide a comparable offering. The fewer viable alternatives for customers, the better a company typically scores for this subfactor. Government regulation can also play a role in the number of competitors.

We also typically assess how a company's product and service offerings compare to those of its peers in terms of unique versus commodity-like offerings. This assessment considers execution and customer service, as well as how technologically advanced the product and service offerings are. We also typically consider trends in price per unit — the price paid for internet capacity, for example — which gives some indication of any price erosion and therefore a sense of the value added and how easily the customer can replace the product or service. A company with a differentiated offering usually scores higher than one with a commodity-like offering. We will take into account if a company does not keep up with evolving technology, which may lead to a lower score for this subfactor.

Companies generally provide some information about the tenure of their contracts and customer churn. For example, satellite operators derive revenue from satellite-distributed video markets and data services. Since video contracts are typically for at least 10 years, compared with about three years for data contracts, we usually consider the revenue mix from different end markets, as well as any available information about individual contracts. Tower operators in markets with developed tower sharing can score higher for this subfactor, because of the limited number of competitors. We may also consider the company's revenue backlog and its churn metrics, if disclosed, to better assess the degree of recurring revenue.

Factor: Profitability and Efficiency (10% weight)

Why it matters

A company's profitability matters because it supports the generation of sustainable cash flow and reflects the strength of the company's competitive position. Profitability also provides insights into a company's ability to manage key costs and pass them on to customers, which is an important indicator of the long-term viability of its business model. Strong profitability and operating efficiency can help a company to withstand economic and cyclical downturns that affect its competitive position and credit metrics.

How we assess it for the scorecard

We score this factor based on one quantitative subfactor: EBITDA Margin.[3]

EBITDA Margin

We use the ratio of earnings before interest, taxes, depreciation and amortization (EBITDA) to revenue.

Factor: Leverage and Coverage (35% weight)

Why it matters

Leverage and debt service coverage provide insight into the level of risk a company is willing to undertake. The degree of leverage is an important indicator of a company's financial flexibility, its ability to invest in growth opportunities and its capacity to pay debt service.

How we assess it for the scorecard

We score this factor based on three quantitative subfactors: (EBITDA – Capex)/Interest Expense; RCF/Net Debt; and Debt/EBITDA.[4]

(EBITDA – Capex) / Interest Expense

We use the ratio of earnings before interest, taxes, depreciation and amortization (EBITDA) minus capital expenditures (capex) to interest expense.

RCF / Net Debt

We use the ratio of retained cash flow (RCF) to net debt.

Debt / EBITDA

We use the ratio of total debt to earnings before interest, taxes, depreciation and amortization (EBITDA).

Factor: Financial Policy (15% weight)

Why it matters

Financial policy is an important rating determinant because it directly affects a company's debt levels, its strategic direction, the risk of adverse changes in financing and capital structure, and its overall credit quality. Financial policy encompasses the tolerance of management and the board for financial risk and their level of commitment to maintaining a strong credit profile.

Financial risk tolerance serves as a guidepost for investment and capital allocation. Liquidity management is an important aspect of financial policy and can provide insight into risk tolerance. Liquidity management is distinct from the level of liquidity, which we discuss in the "Other considerations" section.

Many companies in this sector have historically used acquisitions to spur revenue growth, expand business lines, consolidate market positions and advance cost synergies.

How we assess it for the scorecard

We assess qualitatively a company's targeted credit profile or desired capital structure, and its ability to achieve its targets. We also consider a company's complexity and transparency, its track record of risk and liquidity management, and its public commitments and history of adhering to its commitments in these areas. We also consider management's operating performance and use of cash flow through different phases of economic and industry cycles, as well as how management responds to key events, such as changes in the credit markets and liquidity environment, legal actions, competitive challenges and regulatory pressures.

When considering event risks in the context of scoring financial policy, we assess the likelihood and potential effect of M&A or other types of balance-sheet-transforming events. We assess management's appetite for M&A activity, with a focus on the type of transactions (i.e., core competency or new business) and funding decisions. We also evaluate the frequency and materiality of acquisitions and previous financing choices. A history of debt-financed or credit-transforming acquisitions generally results in a lower score for this factor. We may also consider a company's willingness to sell itself and the negative repercussions that could result, such as through a leveraged buyout transaction.

We also consider a company's and its owners' past record of balancing shareholder returns and debtholders' interests. We are likely to take a negative view in scoring this factor when a company has a track record of favoring shareholder returns at the expense of debtholders.

Other considerations

Ratings may reflect other considerations that are not in the scorecard, usually because the credit importance of these considerations varies widely among companies in this sector or because they may be important only under certain circumstances or for a subset of companies. Such considerations can include financial controls and the quality of financial reporting; corporate legal structure; the quality and experience of management; assessments of corporate governance, as well as environmental and social considerations; exposure to uncertain licensing regimes; and possible government interference in some countries. Regulatory, litigation, liquidity, technology and reputational risk, as well as changes to consumer and business spending patterns, competitor strategies and macroeconomic trends can also be considerations that affect ratings.

The following are examples of other considerations that may be reflected in our ratings and may cause ratings to be different from scorecard-indicated outcomes.

Environmental, social and governance considerations

Where environmental, social and governance (ESG) issues are meaningful for credit profiles, we incorporate them into our ratings analysis in a variety of ways in the application of our sector-specific methodologies. As one part of our overall credit analysis, we consider how ESG considerations could affect the qualitative and quantitative factors and subfactors in the scorecard.

Even where ESG considerations do not affect the measures in the scorecard, or where they cannot be quantified, we incorporate them into our overall analysis of credit drivers that are meaningful to the rating. As a result, we may incorporate these ESG risks qualitatively outside of the scorecard. As part of our ratings analysis, we may establish Issuer Profile Scores (IPSs), which indicate our opinion of the extent to which a given company is exposed to E, S and G risks (incorporating ESG-specific mitigants) or benefits from its exposure to E, S or G. The IPSs are inputs to credit ratings. For more information, see our methodology that describes our general principles for assessing ESG risks.[5]

Liquidity

Liquidity is an important rating consideration for all companies, and extremely weak liquidity can heavily affect ratings in many cases. Our liquidity assessment can be particularly important where there are significant short-term demands on a company's liquidity, for example, in highly seasonal operating environments where working capital needs must be considered. We form an opinion on likely short-term liquidity requirements from the perspective of both sources and uses of cash. For more information, see our liquidity cross-sector methodology.[6]

Management strategy

The quality of management is an important factor supporting a company's credit strength. Assessing the execution of business plans over time can be helpful in assessing management's business strategies, policies and philosophies, and in evaluating management performance relative to that of competitors and our projections. Management's track record of adhering to stated plans, commitments and guidelines provides insight into management's likely future performance, including in stressed situations.

Financial controls

We rely on the accuracy of audited financial statements to assign and monitor ratings in this sector. The quality of financial statements may be influenced by internal controls, including the proper tone at the top, centralized operations, and consistency in accounting policies and procedures. Auditors' reports on the effectiveness of internal controls, auditors' comments in financial reports and unusual restatements of financial statements or delays in regulatory filings may indicate weaknesses in internal controls.

Excess cash balances

Some companies in this sector may maintain cash balances (meaning liquid short-term investments, as well as cash) that far exceed their operating needs. This excess cash can be an important credit consideration.

The amount of cash required to run a business can vary materially based on the regions of operation and the specific subsectors in which a company operates. As a result, we typically assess cash on the balance sheet qualitatively, taking into account a company's track record and its financial and liquidity policies. Cash holdings may be a material mitigant in our scenario analyses of potential acquisitions, share buybacks or special dividends. We assess possible uses of cash, especially for companies that may be acquisitive or under pressure to return cash to shareholders. By contrast, some companies maintain large cash holdings for long periods in excess of

their operating and liquidity needs solely because of conservative financial policies, which provides a stronger indication of an enduring approach that will benefit creditors. Large cash holdings provide some buffers against credit deterioration, and we typically consider cash balances in our analysis of near-term liquidity sources and uses. See the "Liquidity" section for more information.

In accordance with our view of how a company will use its cash, we may use total debt or net debt in our forward-looking view of credit metrics for the Leverage and Coverage factor. We may also cite rating thresholds based on these metrics.

Additional metrics

The metrics included in the scorecard are those that are generally most important in assigning ratings to companies in this sector; however, we may use additional metrics to inform our analysis of specific companies. These additional metrics may be important in our forward-looking view of metrics in the scorecard or in other rating considerations.

Non-wholly owned subsidiaries

Some companies in this sector choose to dilute their equity stake in certain material subsidiaries, such as through an initial public offering or a sale to a private investor, which may, in some cases, negatively affect future financial flexibility. While such transactions may improve cash holdings on a one-off basis, selling minority interests in subsidiaries may have a negative effect on cash flow available to the parent company that consolidated financial statements may not fully reflect. For example, in the case of an equity stake reduction in a subsidiary down to 75%, in the parent's financial statements, the subsidiary's revenue and EBITDA would typically still be consolidated at the group level.

The presence of minority stakes can increase structural subordination and reduce the parent's share of dividend flows from a non-wholly owned subsidiary. While less frequent, the sale of a minority stake may have provisions that further limit the parent's financial flexibility, for instance restrictions on cash pooling with other members of the corporate family, limitations on dividends and distributions, or arms-length business requirements. Minority stakeholders may have seats on the board of the subsidiary.

In many cases, we consider the impact of non-wholly owned subsidiaries qualitatively. For example, we may find that an additional view of financial results, such as analyzing cash flow on a proportional consolidation basis, may be useful to inform our analysis. When equity dilution or structural subordination from non-wholly owned subsidiaries is material and negative, the credit impact is captured in our ratings.

For companies that hold material minority interest stakes (investor companies), consolidated funds from operations typically includes the dividends received from the equity-accounted entity, while none of its debt is consolidated. The investor company's cash flow may be subject to interruption if it is required for the equity-accounted entity's debt service, capital expenditures or other cash needs. The debt of the equity-accounted entity is structurally senior to the investor company's debt. When an equity-accounted entity has a material level of debt, we may estimate one or more of the investor company's financial metrics using proportional consolidation or another approach to inform our analysis of consolidated financials and capture the credit impact of these considerations in our ratings.

Event risk

We also recognize the possibility that an unexpected event could cause a sudden and sharp decline in a company's fundamental creditworthiness, which may cause the assigned rating to be lower than the scorecard-indicated outcome. Event risks — which vary and can range from leveraged recapitalizations to sudden regulatory changes or liabilities from an accident — can overwhelm even a stable, well-capitalized company. Other types of event risks may include M&A, asset sales, spinoffs, shareholder distributions, litigation, natural disasters, pandemics, significant cybercrime events and geopolitical conflicts.

Regulatory considerations

Companies in this sector are subject to varying degrees of regulatory oversight. Effects of these regulations may entail limitations on operations, higher costs, and higher potential for technology disruptions and demand substitution. Regional differences in regulation, implementation or enforcement may advantage or disadvantage a particular company.

Our view of future regulatory requirements is important because it is the basis for our expectations of future financial metrics and of a company's ability to generate sufficient cash flow relative to its debt burden over the medium and longer term.

Parental support

Ownership can provide ratings lift for a particular company if it is owned by a highly rated owner or owners who view it to be of strategic importance. In our analysis of parental support, we consider whether the parent has the financial capacity and strategic incentives to provide support to the company in times of stress or financial need (e.g., a major capital investment or advantaged operating agreement), or if it has done so in the past. Conversely, if the parent puts a high dividend burden on the company, which in turn reduces its flexibility, the ratings will reflect this risk. For more information, see our cross-sector methodology on assessing the impact of financial support from a parent, subsidiary or affiliate in the absence of a guarantee.[7]

Government-related issuers may receive ratings uplift from expected government support. However, for certain companies, government ownership can have a negative effect on the underlying Baseline Credit Assessment (BCA). For example, price controls, onerous taxation and high distributions can have a negative effect on a company's underlying credit profile. For more information on the BCA, see *Rating Symbols and Definitions* and our cross-sector methodology for government-related issuers.[8]

Other institutional support

In some countries, large companies have received government or banking support in the event of financial difficulties because of their overall importance to the functioning of the economy. We consider whether group and banking relationships may provide support when systemically important companies encounter significant financial stress.

As a notable example, some of our corporate ratings in Japan incorporate the government or banking support we expect to be provided, based on historical experience, to large and systemically important companies. Over the years, this support has resulted in lower levels of default for Japanese companies than might otherwise have occurred.

Scorecard-indicated outcome

We arrive at a scorecard-indicated outcome by first mapping the outcome for each factor or subfactor to a numeric score, and then by producing an aggregate numeric score and the equivalent alphanumeric outcome.

Mapping scorecard factors to a numeric score

We map the outcome of each factor or subfactor in the scorecard to a broad Moody's rating category and to a corresponding numeric score. We score qualitative factors and subfactors based on the description by broad rating category in the scorecard. The following exhibit shows the numeric score that corresponds to each broad rating category.

Exhibit 3

Numeric scores for qualitative factors and subfactors

Aaa	Aa	A	Baa	Ba	B	Caa	Ca
1	3	6	9	12	15	18	20

Source: Moody's Ratings

We score quantitative factors and subfactors on a linear continuum. For each metric, the scorecard shows the range by broad rating category. We use the scale in the following exhibit and linear interpolation to convert the metric, based on its placement within the scorecard range, into a numeric score, which may be a fraction. As an illustrative example, if the Baa range for the ratio of revenue to interest is 50x to 100x, the numeric score for a company with a ratio of 99x – relatively strong within this range – would be closer to 7.5, while the numeric score for a company with a ratio of 51x – relatively weak within this range – would be closer to 10.5. In the scorecard footnotes, we define the endpoints of the line (i.e., the metric values that equate to the highest possible numeric score and the lowest possible numeric score).

Exhibit 4

Ranges of numeric scores for quantitative factors and subfactors

Aaa	Aa	A	Baa	Ba	B	Caa	Ca
0.5 - 1.5	1.5 - 4.5	4.5 - 7.5	7.5 - 10.5	10.5 - 13.5	13.5 - 16.5	16.5 - 19.5	19.5 - 20.5

Source: Moody's Ratings

Determining the overall scorecard-indicated outcome

We multiply the numeric score for each subfactor (or each factor, when the factor has no subfactors) by the weight of that subfactor (or factor) and sum the results to produce an aggregate numeric score. We map the aggregate numeric score back to a scorecard-indicated outcome using the ranges in the following exhibit.

Exhibit 5
Scorecard-indicated outcome

Aggregate numeric score	Scorecard-indicated outcome
x ≤ 1.5	Aaa
1.5 < x ≤ 2.5	Aa1
2.5 < x ≤ 3.5	Aa2
3.5 < x ≤ 4.5	Aa3
4.5 < x ≤ 5.5	A1
5.5 < x ≤ 6.5	A2
6.5 < x ≤ 7.5	A3
7.5 < x ≤ 8.5	Baa1
8.5 < x ≤ 9.5	Baa2
9.5 < x ≤ 10.5	Baa3
10.5 < x ≤ 11.5	Ba1
11.5 < x ≤ 12.5	Ba2
12.5 < x ≤ 13.5	Ba3
13.5 < x ≤ 14.5	B1
14.5 < x ≤ 15.5	B2
15.5 < x ≤ 16.5	B3
16.5 < x ≤ 17.5	Caa1
17.5 < x ≤ 18.5	Caa2
18.5 < x ≤ 19.5	Caa3
19.5 < x ≤ 20.5	Ca
x > 20.5	C

Source: Moody's Ratings

For example, a company with an aggregate numeric score of 11.7 would have a Ba2 scorecard-indicated outcome.

In general, the scorecard-indicated outcome is oriented to the senior unsecured rating for investment-grade companies and the corporate family rating (CFR) for speculative-grade companies. For companies that benefit from rating uplift from parental support, government ownership or other institutional support, we consider the underlying credit strength or BCA for comparison to the scorecard-indicated outcome. For more information on the BCA, see *Rating Symbols and Definitions* and our cross-sector methodology for government-related issuers.[9]

Assigning issuer-level and instrument-level ratings

After considering the scorecard-indicated outcome, other considerations and relevant cross-sector methodologies, we typically assign a senior unsecured rating for investment-grade companies and a CFR for speculative-grade companies. For government-related issuers, we assign a BCA.

We may notch individual debt instrument ratings upward or downward from the senior unsecured rating or the CFR to reflect our assessment of differences in expected loss related to an instrument's seniority level and collateral. For more information, see our cross-sector methodologies on loss given default for speculative-grade nonfinancial companies, notching corporate instrument ratings based on differences in security and priority of claim, and assigning short-term ratings.[10]

Key rating assumptions

For information about key rating assumptions that apply to methodologies generally, see *Rating Symbols and Definitions*.[11]

Limitations

In the preceding sections, we have discussed the scorecard factors and other considerations that may be important in assigning ratings. In this section, we discuss limitations that pertain to the scorecard and to the overall methodology.

Limitations of the scorecard

There are various reasons why scorecard-indicated outcomes may not map closely to assigned ratings.

For example, the scorecard may not fully capture credit loss and recovery considerations, which typically become more important as a company gets closer to default. The scorecard is also limited by its upper and lower bounds, causing scorecard-indicated outcomes to be less likely to align with ratings for companies at the upper and lower ends of the rating scale.

The weights of each scorecard factor and subfactor approximate their importance for rating decisions across the sector, but the importance of a particular factor may vary substantially based on an individual company's circumstances.

Factors outside the scorecard, including those discussed in the "Other considerations" section, may be important for ratings, and their relative importance may also vary from company to company. In addition, certain broad methodological considerations described in one or more cross-sector methodologies may be relevant to ratings in this sector.[12] These considerations include how sovereign credit quality affects nonsovereign issuers, the assessment of credit support from other entities, the relative ranking of different classes of debt and hybrid securities, and the assignment of short-term ratings.

We may use the scorecard over various historical or forward-looking periods. Furthermore, in our ratings we often incorporate directional views of risks and mitigants in a qualitative way.

General limitations of the methodology

This methodology does not include an exhaustive description of all the factors we may consider in assigning ratings in this sector. Companies in this sector may face new risks or new combinations of risks, and they may develop new strategies to mitigate risk. We seek to incorporate all material credit considerations in our ratings and to take the most forward-looking perspective that visibility into these risks and mitigants permits.

Ratings reflect our expectations for a company's future performance; however, as the forward horizon lengthens, uncertainty increases and the utility of precise estimates, as scorecard inputs or in other considerations, typically diminishes. Our forward-looking opinions are based on assumptions that may prove, in hindsight, to have been incorrect. Reasons for this could include unanticipated changes in any of the following: the macroeconomic environment, general financial market conditions, industry competition, disruptive technology, or regulatory and legal actions. In any case, predicting the future is subject to substantial uncertainty.

Moody's related publications

Credit ratings are primarily determined through the application of sector methodologies. Certain broad methodological considerations (described in one or more cross-sector methodologies) may also be relevant to the determination of credit ratings of issuers and instruments. A list of sector and cross-sector methodologies can be found here.

Data summarizing the historical robustness and predictive power of credit ratings can be found here.

For more information, see *Rating Symbols and Definitions*, which is available here.

Endnotes

1 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

2 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

3 For definitions of our most common metrics, see our cross-sector methodology that describes our financial statement adjustments for nonfinancial corporations. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

4 For definitions of our most common metrics, see our cross-sector methodology that describes our financial statement adjustments for nonfinancial corporations. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

5 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

6 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

7 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

8 Links to *Rating Symbols and Definitions* and to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

9 Links to *Rating Symbols and Definitions* and to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

10 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

11 A link to *Rating Symbols and Definitions* can be found in the "Moody's related publications" section.

12 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

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